

STAT 7110 Statistical Models and Methods

Exam 1, Fall 2026

Instructions

1. This is an **individual** take-home exam. You may use the textbook, the course slides, your own notes, and R (or other statistical software). You may **not** discuss the exam with, or receive help from, anyone other than the instructor.
2. Show all your works. A numerical answer without supporting work or reasoning may receive little or no credit.
3. Include your code and the relevant output, and put the requested tables and figures in your answers.
4. Unless stated otherwise, use 95% confidence and round final numerical answers to 3 decimal places.
5. Submit a single PDF file to the course website.

Problem 1 (35 points): Lung function in children

The data file **FEV.csv** contains measurements on 654 children aged 3 to 19 years who were seen in the Childhood Respiratory Disease Study in East Boston, Massachusetts (Rosner, data set FEV.DAT). Forced expiratory volume (FEV), the volume of air a person can exhale in the first second of a forced breath, is a standard measure of lung function. The variables are:

Variable	Description	Coding / units
Id	ID number	—
Age	Age	years
FEV	Forced expiratory volume	liters
Hgt	Height	inches
Sex	Sex	0 = female, 1 = male
Smoke	Smoking status	0 = not a current smoker, 1 = current smoker

- (a) **(12 pts)** Separately for nonsmokers and current smokers, compute the sample size, mean, median, standard deviation, Q1, Q3, and IQR of FEV, and report them in one table. Construct side-by-side box plots of FEV by smoking status and use the $1.5 \times \text{IQR}$ rule to identify any potential outliers. Briefly compare the center, spread, and shape of the two distributions.
- (b) **(10 pts)** A reader concludes from (a) that “smoking is associated with better lung function in children.” Compute the mean **age** in each smoking group, then restrict the data to children aged **12 years or older** and recompute the mean FEV by smoking status. Use these results to explain why the reader’s conclusion is misleading.
- (c) **(13 pts)** Treat the 65 current smokers as a random sample from a population of child smokers. Construct a 95% confidence interval for the population mean FEV, μ , stating which distribution you used and why, and interpret the interval. Then construct a 95% confidence interval for the population standard deviation σ , and state the assumption this interval requires.

Problem 2 (30 points): Evaluating a screening biomarker

A hospital is evaluating a new blood biomarker to detect early-stage hepatocellular carcinoma (HCC, a type of liver cancer). In a validation study the investigators recruited **200** patients with biopsy-confirmed early HCC (cases) and **800** patients with cirrhosis but no HCC (controls). Each patient's biomarker result was reported on an ordinal 1 - 5 scale, where higher scores indicate greater suspicion of cancer. (The data are hypothetical.)

Biomarker score	1	2	3	4	5	Total
HCC cases	6	14	30	60	90	200
Controls (no HCC)	360	240	120	56	24	800

- (a) **(8 pts)** Suppose a score of **4 or higher** is called “test positive.” Construct the 2×2 table of test result by disease status and compute the sensitivity and specificity. A colleague computes $150/230 = 0.652$ and calls it the positive predictive value (PPV). Explain why this number should **not** be reported as the PPV for patients in practice.
- (b) **(12 pts)** In a surveillance program for patients with cirrhosis, the prevalence of undetected early HCC is 3%. Using the “score ≥ 4 ” rule, use Bayes’ rule to compute the PPV and the negative predictive value (NPV). Then compute the PPV in a referral clinic where the prevalence is 20%, and explain why the two PPVs differ even though the test is the same.
- (c) **(10 pts)** Compute the sensitivity and specificity for each cutoff “score $\geq k$,” $k = 1, \dots, 5$. Plot the ROC curve, compute the area under the curve (AUC) with the trapezoidal rule, and interpret the AUC in words.

Problem 3 (35 points): Infection surveillance and costs

A hospital infection-control team monitors central line–associated bloodstream infections (CLABSIs) in its intensive care units (ICUs). Under baseline conditions, CLABSIs occur at a rate of **1.5 infections per 1,000 catheter-days**, and a typical ICU accumulates **2,400 catheter-days per quarter** (3 months). Let X be the number of CLABSIs in one ICU in one quarter, and assume X follows a Poisson distribution.

- (a) **(12 pts)** State the distribution of X and its mean. Find $P(X = 0)$ and $P(X \geq 8)$. One ICU reports 8 CLABSIs this quarter; is this unusual under baseline conditions? State one assumption of the Poisson model that might be questionable in this setting.
- (b) **(11 pts)** The hospital has 10 such ICUs, each operating independently under baseline conditions, and it “flags” an ICU for review if it has 8 or more CLABSIs in a quarter. Let Y be the number of ICUs flagged in a quarter. State the distribution of Y , and find $E(Y)$ and $P(Y \geq 1)$. What does $P(Y \geq 1)$ imply about the flagging policy?
- (c) **(12 pts)** Each CLABSI adds an excess hospital cost with mean \$46,000 and standard deviation \$30,000; individual costs are strongly right-skewed. Suppose next year there will be 50 CLABSIs with independent costs, and let $\bar{C}$ be the average excess cost per infection. Find the mean and standard deviation of $\bar{C}$, approximate $P(\bar{C} > \$52,000)$, and explain why a normal approximation is reasonable even though individual costs are skewed.